

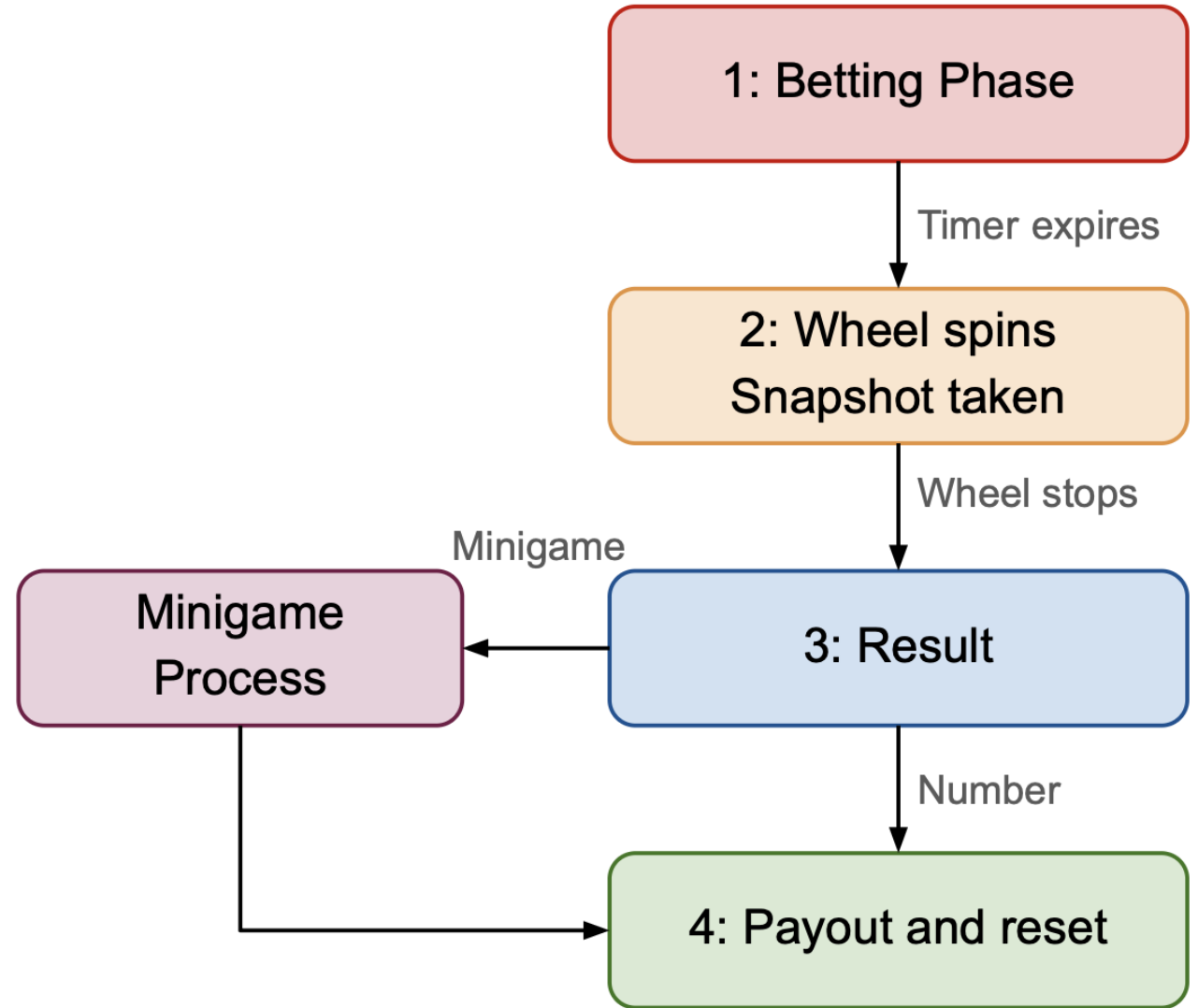


Distributed Crazy Time

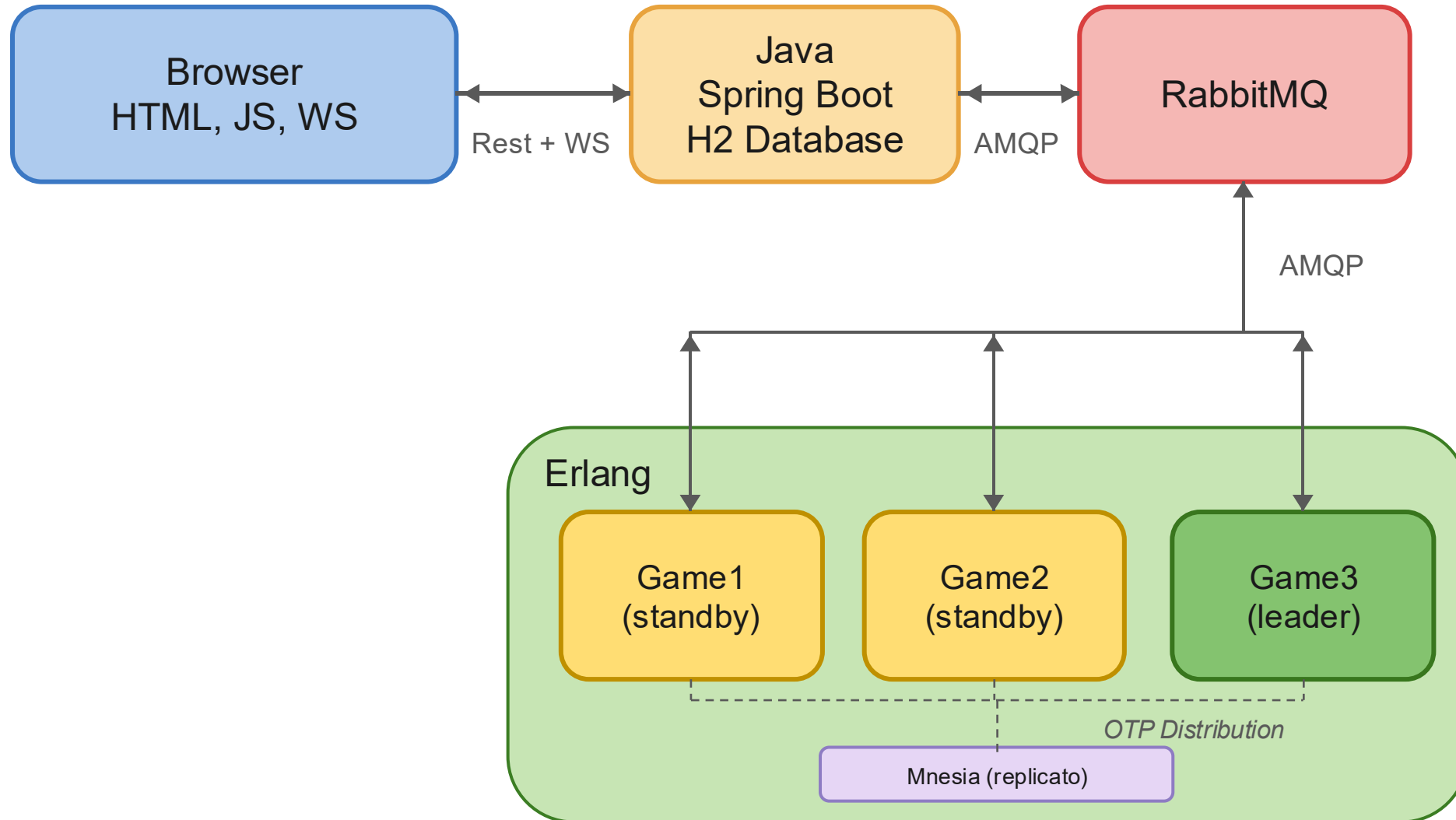
Live distributed betting system with fault tolerance

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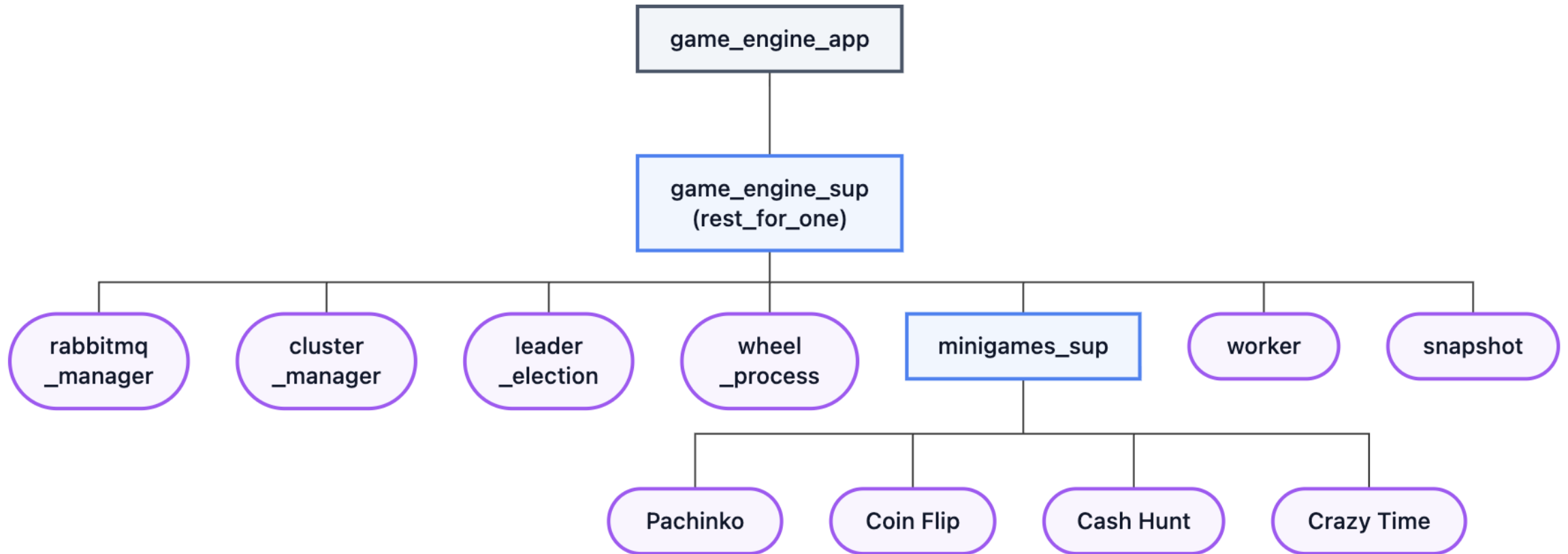
Game Loop



System Architecture



Supervision Tree



Cluster manager

Reads peer_nodes from config file

Monitors with net_kernel:monitor_nodes/2

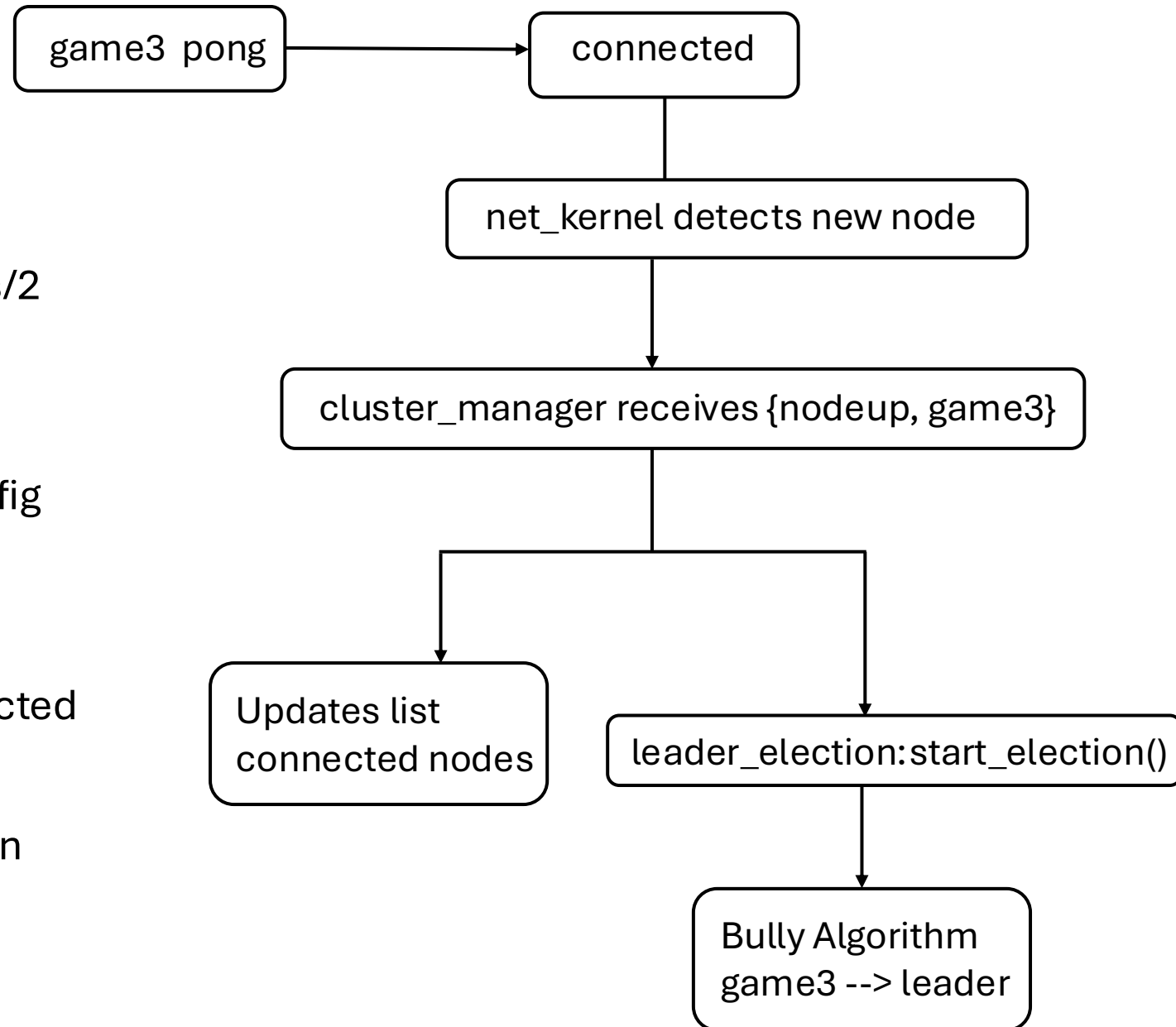
At boot:

1. Pings every peer in the peer_node config
2. Starts leader_election
3. Bootstrap Mnesiadb
4. Ping every 10s for peers not yet connected

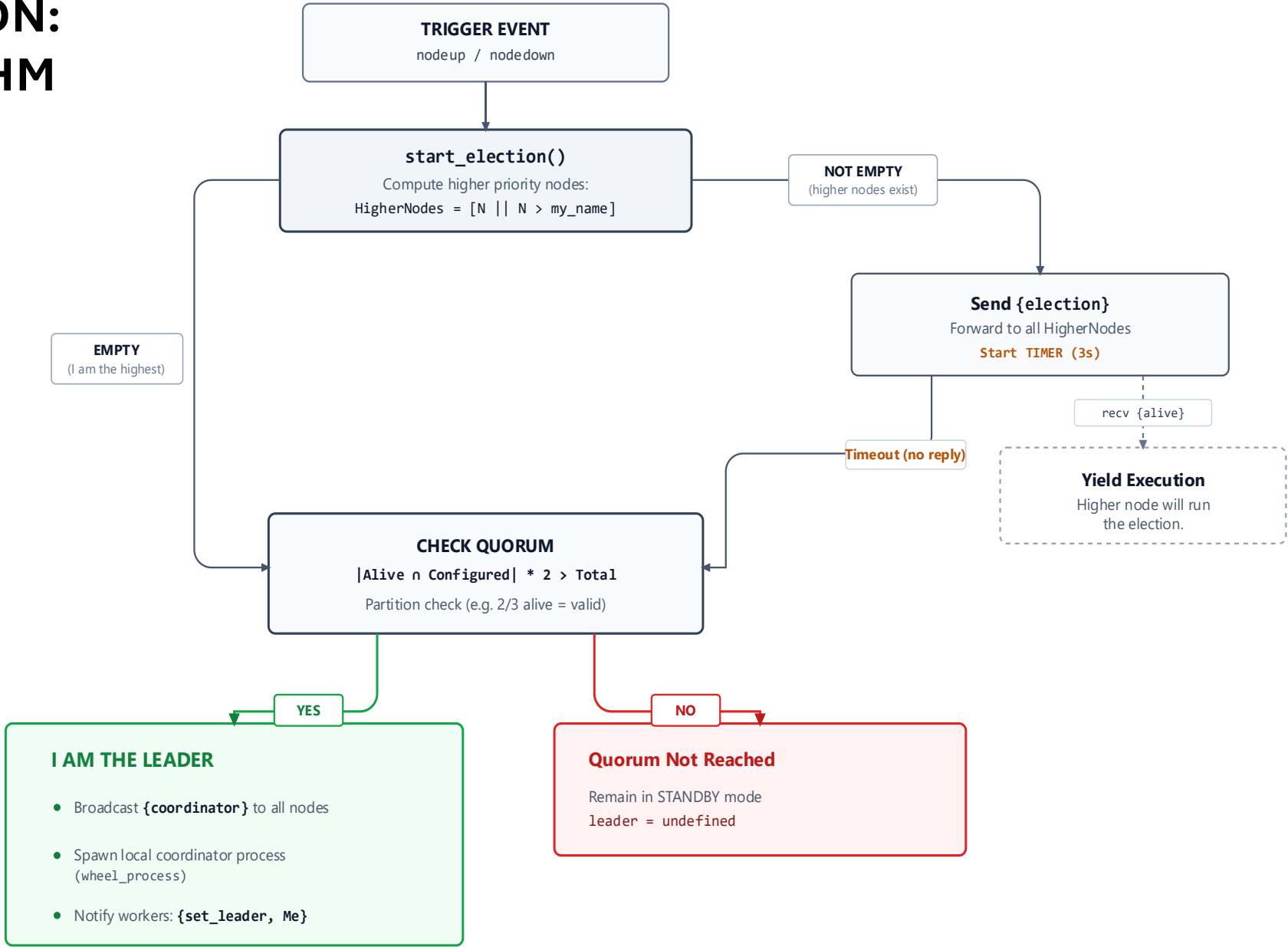
Node_up / Node_down -> leader_election

Mnesiadb exists -> join_mnesiadb()

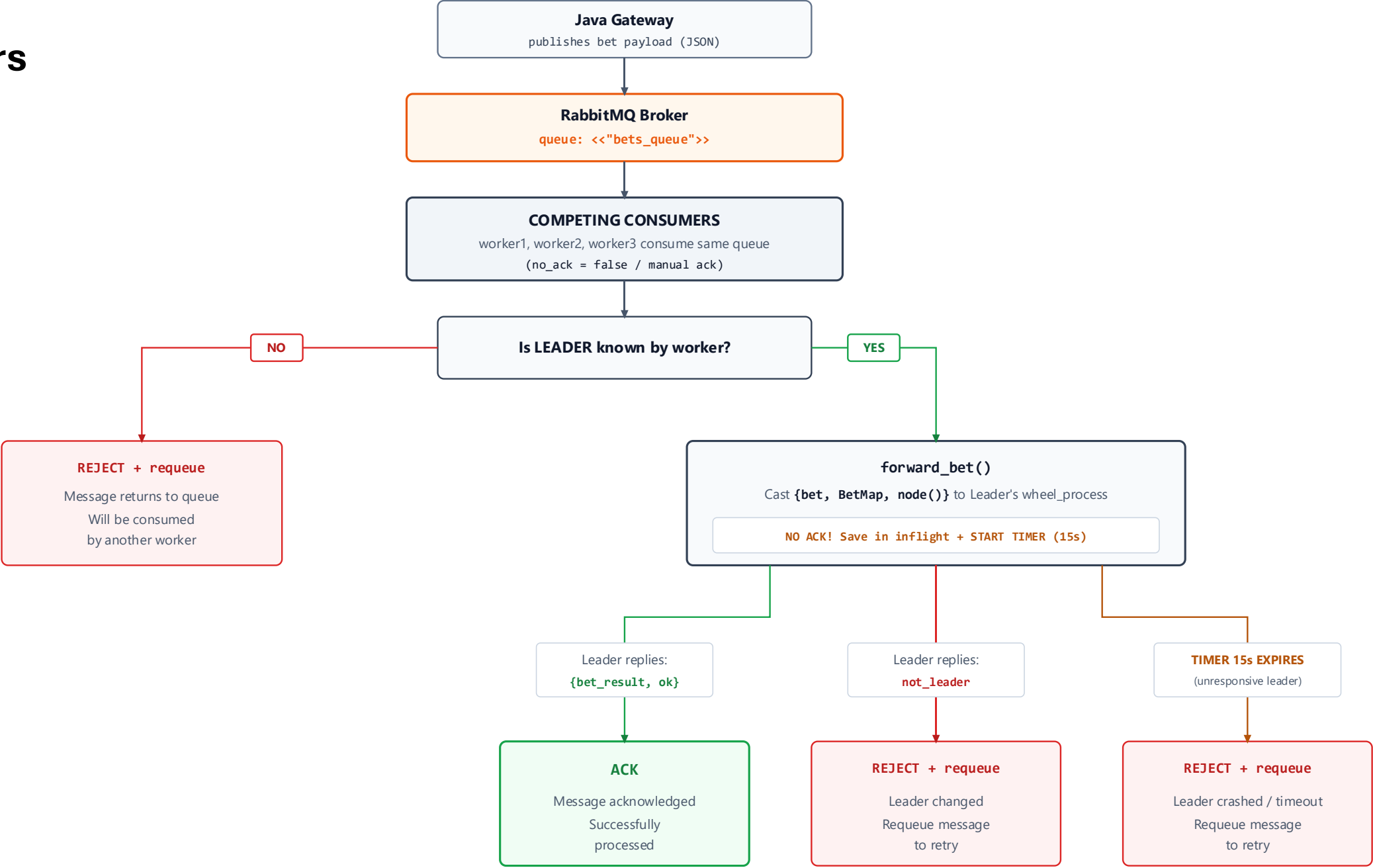
Otherwise the lowest node will create it



LEADER ELECTION: BULLY ALGORITHM

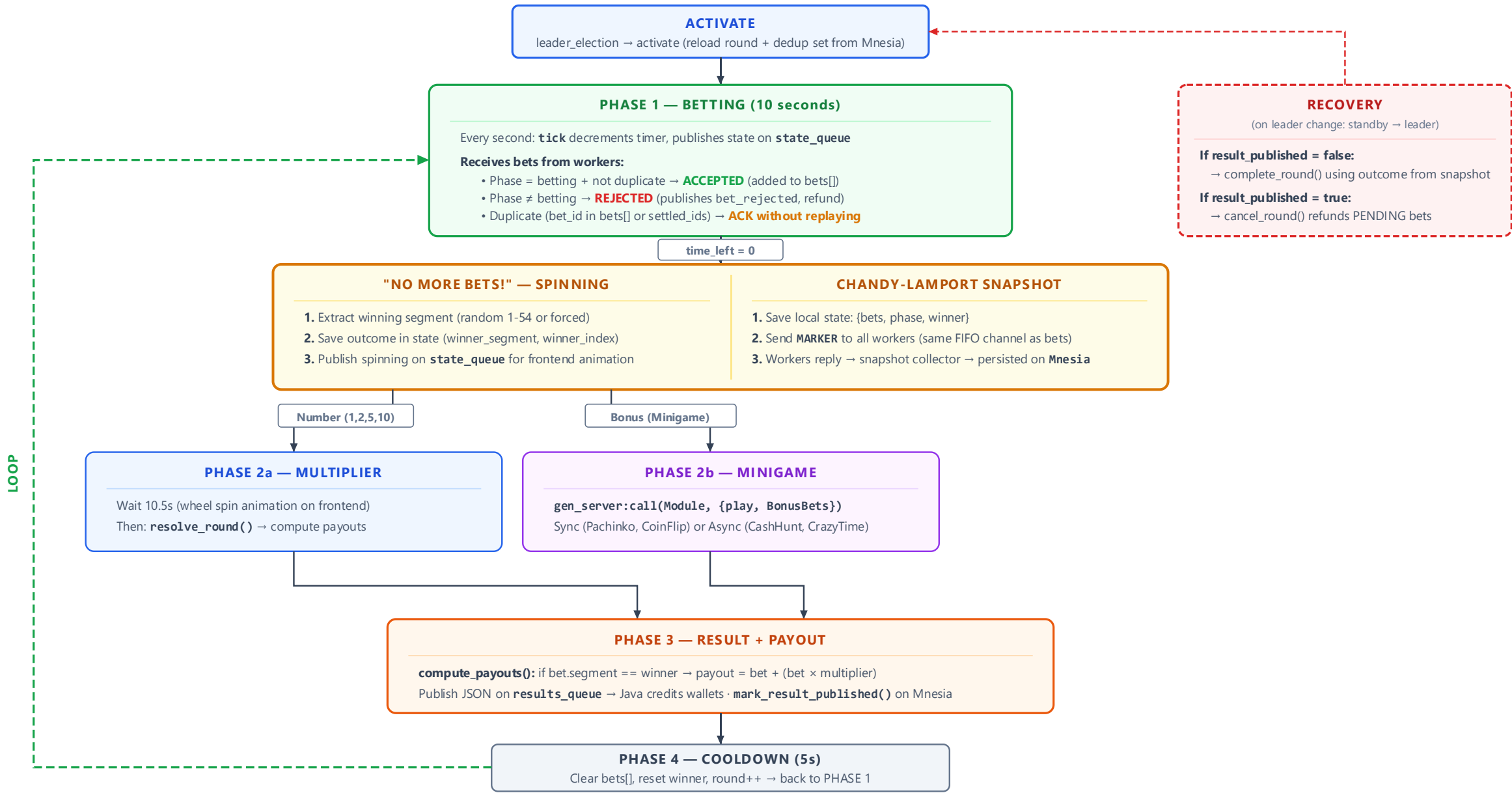


Workers



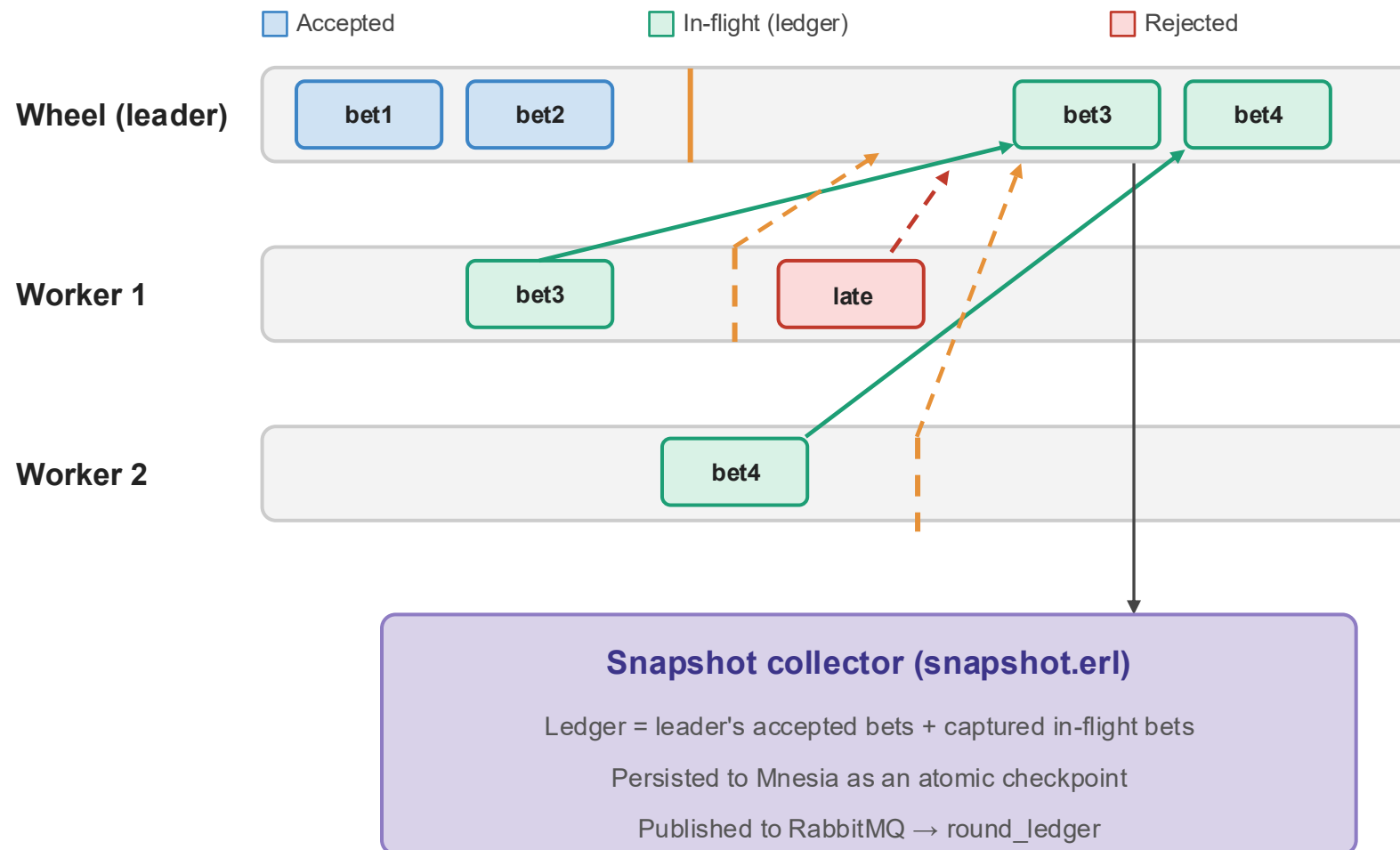
Wheel Process — Game Loop

Runs ONLY on the Leader node — standby nodes have the wheel deactivated



Distributed Snapshot — Chandy-Lamport Algorithm

Cut points are staggered — the marker arrives at each node at a different time.



- The marker travels the same FIFO channel as the bets, so causal order is guaranteed.
- A bet forwarded before the marker is in-flight — captured if it lands before the marker returns to the leader.
- A bet sent after the marker on that channel arrives too late and is rejected.

Chandy-Lamport Snapshot: How and Why

When Triggers at the Gong (end of betting), right after the winning segment is drawn, before the wheel animation starts.

How: 3-Phase Protocol

1. Wheel (Leader)

Saves its own state, starts the **collector**, and sends **markers** to the workers on the same FIFO channels as the bets.

2. Worker

Each worker saves its local state, replies with the **marker**, and reports its portion back to the collector.

3. Collector

Assembles the `snapshot_record` (local bets + in-flight), persists it to **Mnesia** (replicated), and publishes the ledger to `results_queue`.

Reasons

Authoritative Ledger

The **definitive** list of the round's bets, including those still **in flight** on the channels.

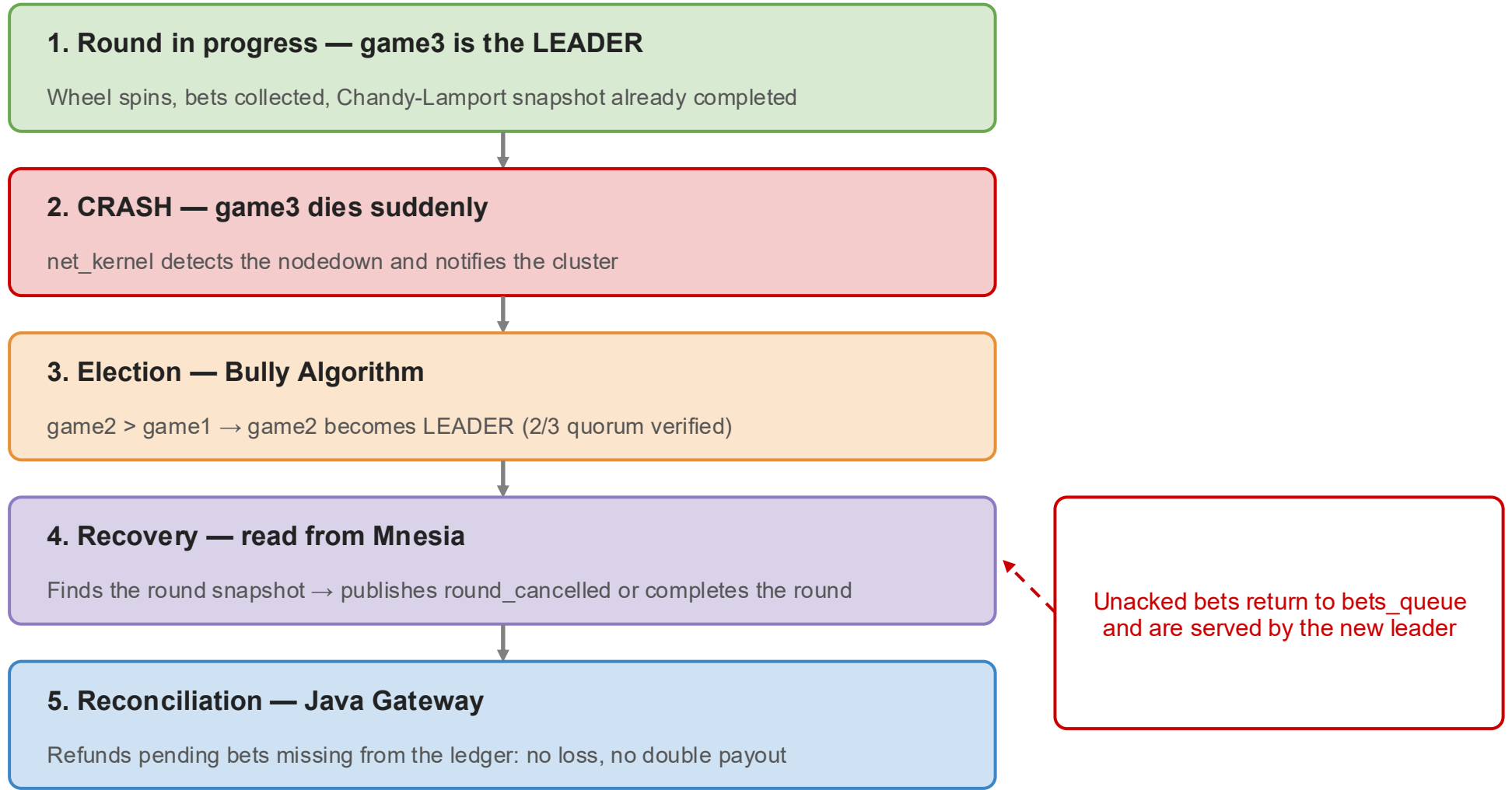
Deterministic Recovery

A new leader reads the checkpoint and decides whether to **complete** (post-gong crash) or **cancel** (betting-phase crash) the round.

Deduplication

The `bet_id` values in the checkpoint prevent bets re-delivered by the broker from being played **twice**.

What Happens When the Leader Crashes?



The Role of the Java Gateway

Transactional Handling

```
POST /api/auth/register
POST /api/auth/login
GET  /api/wallet/balance
POST /api/wallet/place-bet
```

place-bet() does:

1. Deducts the balance (pessimistic lock)
2. Saves the bet as PENDING
3. Sends the JSON to bets_queue

Reconciliation from the Ledger

From RabbitMQ → results_queue

type: "result"

PayoutListener: computes winnings, updates status (WON/LOST), credits the balance

type: "round_ledger"

LedgerListener: PENDING bets missing from the ledger → REFUNDED

type: "round_cancelled"

LedgerListener: refunds ALL PENDING bets of the round

Testing

Automated verification: load handling and dealer-crash recovery

STRESS TEST

stress_test.py

- 50 registered players run an open-ended sequence of concurrent rounds
- Each round: a random amount is wagered on a random segment, for every player, in parallel
- Interactive checkpoints — the operator presses Enter when betting opens and again once the wheel stops
- Per round it prints, for every player: balance before/after, payout, net result — plus the house balance

Goal: verify correctness and responsiveness of the whole pipeline under concurrent load

LEADER CRASH TEST

crash_test.py

1. Setup

10 players each wager 10 EUR on segment 5 — the amount actually debited is recorded

↓ *moments before the gong*

2. Simulated crash

pkill -9 is issued against game3 (the Leader), before the gong

↓ *Bully election + Mnesia-checkpoint recovery run automatically*

3. Monitoring

Player balances are sampled every 5s for one minute

↓ *Java Gateway reconciles the ledger*

4. Verification

Refunded total vs. staked total → confirms which recovery branch ran and that no money was lost

Both suites confirm zero financial loss: correct concurrency handling under load, and correct Bully election + Chandy-Lamport recovery after a dealer crash.